

FreshGuard IMCS

Slide-by-Slide Presentation Script

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How to use this script

Speak naturally. Do not read every word too fast. The timing targets keep the full presentation close to 15 minutes.

Overall timing

16 slides at roughly 45 to 70 seconds each. Aim for about 14 to 15 minutes total.

1. FreshGuard IMCS

Timing target

45 seconds

Main message

Introduce the project and frame it as an honest fruit-quality inspection system.

What to say

Good morning. My name is Abrar Mustafa, and my student ID is 925259. My project is called FreshGuard IMCS, which stands for a leakage-aware fruit quality inspection system. The project uses computer vision to classify fruit images as Good or Bad quality, and it also predicts the specific fruit-quality class. The main point of this project is not only to achieve high accuracy. The main point is to make sure the accuracy is trustworthy, because the dataset contains many near-duplicate fruit images.

Transition

To understand why this matters, I will first explain the problem the project is solving.

2. The Project Story

Timing target

50 seconds

Main message

Explain the problem, risk, and response in simple language.

What to say

The problem is that fruit quality inspection needs a reliable way to separate Good fruit from Bad fruit before shipping. If a bad fruit is accepted as good, it can reach the customer, which is a serious quality issue. The risk in this dataset is that a normal image split can place almost identical fruit photos in both training and testing. If that happens, the model may look very accurate because it has already seen a very similar image before. FreshGuard responds to this by checking for near-duplicate leakage first, then creating a clean split before training and testing models.

Transition

Next, I will show what data the project uses and how the dataset is structured.

3. Dataset and Scope

Timing target

55 seconds

Main message

Describe the FruitNet dataset and the balanced 12-class setup.

What to say

The project uses the FruitNet dataset, which contains Indian fruit images with quality labels. I converted the dataset into 12 classes. There are six fruit types: Apple, Banana, Guava, Lime, Orange, and Pomegranate. Each fruit has two quality labels: Good and Bad. To keep training balanced, every class is capped at 600 images. So the final working dataset contains 12 classes and 7,200 images in total. This balanced setup is important because it prevents one class from dominating the model during training.

Transition

After building the dataset, I checked the images visually to make sure the labels made sense.

4. Visual Sanity Check

Timing target

45 seconds

Main message

Explain why visual checking matters before model training.

What to say

Before training any model, I performed a visual sanity check. This means I displayed one Good and one Bad image for each fruit type. This step is simple, but it is very important. It helps confirm that the images are readable, the folder labels are correct, and the Bad fruit examples actually show visible defects such as bruises, rot, marks, or discoloration. This gives confidence that the model will be trained on meaningful labels.

Transition

Now I will explain the biggest issue in the project: leakage in a naive train/test split.

5. Why Naive Accuracy Can Be Misleading

Timing target

65 seconds

Main message

Make the leakage problem clear and memorable.

What to say

A normal image-level split divides images into training, validation, and test sets. At first, this sounds fine. But in this dataset, many images are near-duplicates. That means the same physical fruit, or an almost identical photo, can appear in both training and testing. The notebook found that 619 out of 1,080 naive test images had near-duplicate leakage. That is 57.31 percent of the test set. This is a major issue because the model may memorize the same fruit instead of learning general quality patterns. So a high test accuracy from this naive split would not be fully reliable.

Transition

To fix this, the project uses a leakage-aware split.

6. Leakage-Aware Split

Timing target

60 seconds

Main message

Summarize the fix: hash, group, split, re-audit.

What to say

The leakage-aware split has four main steps. First, every image is converted into a perceptual hash, which is like a compact fingerprint of the image. Second, visually similar images are grouped together by comparing their hash distances. Third, whole groups are assigned to train, validation, or test. This means that near-duplicate images stay in the same split and do not cross between training and testing. Finally, the clean split is audited again. After this process, the test leakage becomes 0.00 percent, which makes the final evaluation much more honest.

Transition

Let me now show the complete pipeline from start to finish.

7. End-to-End Pipeline

Timing target

60 seconds

Main message

Walk through the whole project flow at a high level.

What to say

The full pipeline has six stages. First, I collect the dataset by downloading FruitNet and storing it in Google Drive. Second, I clean the data by keeping Good and Bad folders and removing ambiguous Mixed labels. Third, I balance the dataset by creating 12 classes with 600 images each. Fourth, I audit near-duplicate leakage before trusting any model score. Fifth, I train two pretrained models: ResNet-18 and ViT-Base/16. Finally, I explain the model decisions using Grad-CAM and LIME. So the pipeline is built around data quality first, then model training, then trust and deployment evidence.

Transition

Now I will go deeper into how the dataset was prepared.

8. Dataset Preparation Details

Timing target

55 seconds

Main message

Explain folder standardization, balancing, and label discipline.

What to say

The original FruitNet folders were converted into a simple PyTorch ImageFolder structure. This means each class has its own folder, such as Apple_Bad, Apple_Good, Banana_Bad, and Banana_Good. This makes the dataset easier to load and train on. The project also uses balanced sampling, with 600 images in every class. Another important decision is label discipline. Mixed-quality folders are removed because they are ambiguous. If the labels are unclear, the model may learn confusing patterns. So the dataset preparation step improves both fairness and label quality.

Transition

The next step is the leakage audit itself, which is the key method in this project.

9. Leakage Audit Mechanics

Timing target

70 seconds

Main message

Explain dHash and Hamming distance without sounding too technical.

What to say

The leakage audit uses a method called perceptual hashing. In the notebook, each image is converted into a 64-bit difference hash, or dHash. A dHash is a small fingerprint that captures the visual pattern of an image. If two images look very similar, their hashes will also be similar. To compare two hashes, the notebook uses Hamming distance, which counts how many bits are different. A smaller distance means the images are more similar. In this project, images are treated as near-duplicates when the Hamming distance is less than or equal to 5. This method found 3,472 near-duplicate groups, and the largest group had 417 images. That proves why a normal split is risky.

Transition

After finding the near-duplicates, the project creates a clean split strategy.

10. Clean Split Strategy

Timing target

60 seconds

Main message

Compare naive and group-aware splitting.

What to say

The clean split is created at group level instead of image level. In the naive split, the train, validation, and test sets were class-balanced, but near-duplicate leakage remained. The naive test set had 57.31 percent leakage. In the group-aware split, all near-duplicate images stay together in one split. The final clean split has 5,213 training images, 904 validation images, and 1,083 test images. After re-auditing the clean split, the leakage is 0 out of 1,083 test images. This makes the final test result much more meaningful.

Transition

With the clean split ready, the notebook moves into model training.

11. Modeling Pipeline

Timing target

60 seconds

Main message

Explain preprocessing, models, evaluation, and explainability.

What to say

The modeling pipeline starts with preprocessing. Images are resized to 224 by 224 pixels and normalized using ImageNet statistics because both models are pretrained on ImageNet. For training, the notebook also uses augmentation such as horizontal flip, small rotation, and mild color jitter. Then two models are trained on the clean split: ResNet-18 and ViT-Base/16. The models are evaluated using multiclass accuracy, macro-F1, confusion matrices, and binary Good/Bad monitoring. Finally, Grad-CAM and LIME are used to check whether the models focus on meaningful fruit defect regions.

Transition

Now let us compare the training results and final model performance.

12. Training and Final Model Comparison

Timing target

65 seconds

Main message

Present ResNet-18 as the stronger practical candidate.

What to say

ResNet-18 is trained for 5 epochs, and ViT-Base/16 is trained for 3 epochs. On the clean test set, ResNet-18 achieves 99.35 percent accuracy and 0.9930 macro-F1. ViT-Base/16 achieves 98.34 percent accuracy and 0.9801 macro-F1. Both models perform very well, but ResNet-18 is slightly better in this experiment. It is also much smaller, with 11.2 million parameters compared with 85.8 million parameters for ViT-Base/16. Because of this, ResNet-18 is the stronger deployment candidate.

Transition

Accuracy is useful, but the confusion matrix gives a more detailed view of mistakes.

13. Confusion Matrix Evidence

Timing target

55 seconds

Main message

Explain what the confusion matrix proves.

What to say

A confusion matrix shows which classes the model predicts correctly and which classes it confuses. A good confusion matrix is mostly diagonal. In the ResNet-18 confusion matrix, most predictions are on the diagonal, which means the model usually predicts the correct fruit-quality class. There are only a small number of off-diagonal mistakes. This is important because accuracy gives one overall number, but the confusion matrix shows the class-level behavior. It helps us see that the model is not only accurate overall, but also consistent across the fruit classes.

Transition

For real inspection, the project also evaluates a simpler Good/Bad decision.

14. Production View: Good/Bad Monitoring

Timing target

70 seconds

Main message

Explain false accept and why it matters most.

What to say

In production, the most important decision may not be the exact fruit class. The main decision is often whether to accept or reject the fruit. So the notebook also converts the 12 classes into a binary Good/Bad monitoring task. For ResNet-18, the binary accuracy is 99.82 percent. The false-accept rate is 0.37 percent, and the false-reject rate is 0.00 percent. False accept means Bad fruit is predicted as Good. This is the most important error to reduce because it means defective fruit may be shipped. ResNet-18 has the lower false-accept rate, so it is better for this production-style monitoring goal.

Transition

Finally, I will explain how the project checks model trust using explainability.

15. Explainability and Trust

Timing target

70 seconds

Main message

Explain Grad-CAM, LIME, and t-SNE in plain language.

What to say

The project uses explainability to check whether the model is looking at the right parts of the image. For ResNet-18, Grad-CAM highlights the image regions that strongly influenced the prediction. For ViT-Base/16, LIME perturbs image regions and shows which superpixels are important for the prediction. In the example, the highlighted areas are mainly around the fruit body and defect region, which supports the idea that the model is using visual quality cues. The project also uses t-SNE to visualize ViT embeddings. The clusters show that the model learned useful representations for separating fruit-quality classes. This makes the result more trustworthy than accuracy alone.

Transition

I will close with the main conclusions and future work.

16. Conclusion

Timing target

65 seconds

Main message

End with the honest-evaluation message.

What to say

To conclude, FreshGuard IMCS is a complete fruit-quality inspection pipeline. It starts with dataset download and cleaning, builds a balanced 12-class dataset, checks visual quality, detects leakage, fixes the split, trains two pretrained models, evaluates them carefully, and adds explainability. The most important finding is that a naive split had 57.31 percent near-duplicate leakage, so the project needed a cleaner evaluation method. After group-aware splitting, leakage became 0.00 percent. On this clean split, ResNet-18 performed best and is also smaller, making it a strong deployment candidate. For future work, the project can test MobileNet or EfficientNet, apply quantization, tune thresholds to reduce false accepts, and test on fresh real-world images. Thank you.

Transition

End of presentation.